Jacob Pieniazek

Lead Data Scientist | Causal Inference & Technical Leadership

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Experience

84.51° (KROGER) | Lead Data Scientist

Feb 2026 – Present | Cincinnati, OH

- ➔ Set scientific direction and lead technical delivery for five data scientists measuring 1,000+ campaigns annually. Experimental and quasi-experimental findings inform dozens of Kroger Marketing stakeholders.
- ➔ Lead a two-scientist causal ML R&D team with weekly household scoring across eight campaign types. Targeting consistently delivers roughly 2x the causal lift of business-as-usual targeting across production and validated use cases. Translate non-linear CATE patterns into marketing insights.
- ➔ Negotiated, designed, and led a 100K-household randomized audit of a partner paid on incremental profit. Found partner-reported lift was roughly twice the experimental estimate and recalibrated profit-sharing payments, preventing significant overpayment.
- ➔ Deliver millions in annual cloud/platform savings by modernizing infrastructure, right-sizing Databricks compute, and optimizing PySpark pipeline execution.
- ➔ Established Claude Code-assisted development practices and AI-generated CI summaries for the development team.
- ➔ Develop LangChain-based agentic pipelines for econometric reporting and insights.
- ➔ Mentor four new hires and lead Miami University data science recruiting. Co-lead a biweekly causal inference learning group. Designed interview cases and rubrics and co-built internal causal inference training for scientists and business teams.

84.51° | Senior Data Scientist

Jul 2024 – Feb 2026 | Cincinnati, OH

- ➔ Architected a Python measurement platform using Kedro and shared Ibis transformations across local DuckDB and production PySpark. Migrated Azure Data Factory workflows to GitHub Actions and Databricks, cutting pipeline costs by 50%.
- ➔ Measured roughly $120M in full-year FY2024 incremental sales for a personalization program spanning 30+ campaigns and 20M households. Designed and deployed its experimental framework with permanent household treatment assignments and a common allocation ratio across entry cohorts to capture cumulative program effects.
- ➔ Integrated Double Machine Learning into program-level measurement with cross-fitted estimation and randomized placebo tests to check for spurious lift.

84.51° | Data Scientist

Jun 2022 – Jul 2024 | Cincinnati, OH

- ➔ Corrected email-suppression bias using offer-specific intent-to-treat or instrumental-variable estimators and added automated integrity checks to production campaign measurement.
- ➔ Integrated causal and econometric methods into measurement workflows and internal documentation.

Open Source & Personal Projects

CAML | Causal ML & AutoML (docs) (code)

- ➔ Building an open-source Python framework to automate selection and validation of heterogeneous treatment-effect models using EconML, FLAML, and Optuna.

NETBEAT | Network Diagnostics CLI (crates) (code)

- ➔ Built a Rust client/server CLI for throughput and latency diagnostics, with concurrent TCP connections and JSON reports for automation.

SELF-HOSTED AI PLATFORM | Personal LLM Infrastructure

- ➔ Build and operate a personal AI platform with centralized cloud/local model routing (LiteLLM), GPU-backed inference, managed multi-user access, agent workflows, persistent memory, and semantic retrieval.

Skills

CORE STRENGTHS

- Technical Leadership
- Causal ML & Experimentation
- Applied Econometrics
- Data Engineering
- MLOps, DevOps, CI/CD
- Production & AI Systems

PROGRAMMING

Python • SQL • Rust
Bash • R • Stata

ML / TOOLS

scikit-learn • PyTorch • XGBoost
EconML • DoubleML • CausalML
FLAML • Optuna • MLflow

AI / AGENTS

LangChain • MCP
Claude Code • LiteLLM
Embeddings
Agentic Development

SYSTEMS / MLOPS

Databricks • PySpark • Delta
Kedro • Ibis • DuckDB
Azure • Terraform
GitHub Actions • FastAPI
Linux • Docker • Kubernetes
PostgreSQL • pytest

Education

MIAMI UNIVERSITY

M.A. Economics | GPA: 4.0
2022 | Full Graduate Assistantship

UNIVERSITY OF DAYTON

B.S. Business Economics & Applied Mathematical Economics
Double Major | 2021 | GPA: 3.92
Summa Cum Laude, Honors Diploma
E. B. O'Leary Award of Excellence

INVITED TALKS

Causal Machine Learning in Practice
Miami University Economics Brown Bag 2026

Interests/Hobbies

Open-Source Development • Self-Hosting / Homelab • Home Automation • Hiking / Camping • Guitar • Audiophile Pursuits